

Anti-Scam: User-Centric Evaluation of LLM-Powered and Content-Based Phone Scam Detection

Mafaz Alhulifi
Taibah University
Madinah, Saudi Arabia
malhulifi@taibahu.edu.sa

Raed Alharbi
Prince Sattam Bin Abdulaziz
University
Riyadh, Saudi Arabia
rf.alharbi@psau.edu.sa

Ashwaq Alsoubai
King AbdulAziz University
Jeddah, Saudi Arabia
atalsoubai@kau.edu.sa

Abstract

Abstract: Phone scams remain a critical threat to both telecom operators and end users, inflicting substantial financial and psychological harm. Traditional defenses—largely based on blacklisting known scam numbers—are quickly outpaced by attackers leveraging VoIP to cycle through fresh identities. In this work, we present Anti-Scam, a content-based detection framework powered by large language models (LLMs) to analyze live call transcripts and flag fraudulent content in real time. Unlike closed, proprietary systems, our tool is fully open for public evaluation and incorporates a human-in-the-loop mechanism: users can review flagged calls, feeding those labels back into the model to continuously refine its accuracy and mitigate keyword or context biases. By combining AI analysis with community-driven oversight, our system offers a robust, adaptable defense against evolving phone-based scams.

CCS Concepts

• **Computing methodologies** → **Natural language processing**;
• **Human-centered computing** → **Collaborative and social computing**; • **Security and privacy** → **Human and societal aspects of security and privacy**.

Keywords

phone scams, large language models, scam detection, content analysis, user feedback, real-time protection

ACM Reference Format:

Mafaz Alhulifi, Raed Alharbi, and Ashwaq Alsoubai. 2025. Anti-Scam: User-Centric Evaluation of LLM-Powered and Content-Based Phone Scam Detection. In *Companion of the Computer-Supported Cooperative Work and Social Computing (CSCW Companion '25)*, October 18–22, 2025, Bergen, Norway. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3715070.3757231>

1 Introduction

The evolution of mobile communication has profoundly transformed global communication by connecting billions of mobile devices and making real-time interactions more accessible than ever before [22]. Yet, this unprecedented level of connectivity has

introduced a range of new vulnerabilities, with phone scams emerging as one of the most pervasive and damaging threats [1]. In 2024, phone scams continue to pose a significant global challenge, with cyber-enabled fraud accounting for nearly 83% of all financial losses reported to the FBI’s Internet Crime Complaint Center [6]. Beyond measurable financial losses, the psychological toll of phone scams is equally alarming, with victims often experiencing a profound loss of trust, both in technology and in their personal interactions, which can lead to increased anxiety, depression, feelings of shame, and reduced self-esteem [5, 12].

Therefore, protecting end-users from phone scams has become a critical priority for policymakers [2–4, 7], technology developers [10, 14], and socio-computing researchers [9, 18, 19]. While these emerging technologies aim to offer more effective protection for users, a significant challenge remains as end-users often cannot evaluate the effectiveness of these tools, as they are typically proprietary and operate behind corporate walls or are only accessible through research publications. To address this gap, we present our large language models (LLMs)-based phone scam detection tool, which will be openly accessible for public evaluation. It is important to clarify that our system currently focuses on detecting scams in software-based voice calls, rather than traditional cellular phone calls. Unlike closed systems, our tool integrates a human-in-the-loop feature that allows users to contribute to the ongoing refinement of the LLM model, enabling continuous improvement and transparency of scam prevention technologies over time. This approach also aligns with the core values of CSCW’s research strides: participatory design, human-centered computing, and collaborative innovation. Visitors to our demo session will gain hands-on experience with our system that showcases how end-users can be integrated into the development and evaluation of cutting-edge LLM and content-based phone scam detection solution.

2 Gaps In Existing Scam Detection Systems

Most prior works on detecting phone scam calls have focused on blacklisting techniques to identify malicious calls [8, 11, 16]. These approaches aim to classify and block scammers’ phone numbers using aggregated user reports and number-based patterns [11, 23]. Although many strides in research have been conducted to advance blacklisting systems, achieving more accuracy than 75% [20], these methods have been found to remain vulnerable to evolving fraudulent tactics, such as caller ID spoofing and number rotation [19, 23]. Thus, researchers have shifted their focus to leveraging patterns in scam calls, such as call duration, frequency, geographic location, call rate, caller behavior, and reputation, for phone scam detection [13, 15, 21]. Although this approach has improved the detection

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CSCW Companion '25, Bergen, Norway

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 979-8-4007-1480-1/25/10

<https://doi.org/10.1145/3715070.3757231>

accuracy of phone scams compared to traditional blacklisting, it still faces challenges with false positives, particularly when legitimate calls follow patterns similar to malicious ones [13, 15, 21]. These recurring challenges highlight that, despite ongoing efforts in this area, a significant gap remains in developing detection approaches that can effectively adapt to the evolving tactics of scammers, which has led researchers to present content-based detection approaches that analyze the actual information exchanged during calls to enhance detection accuracy [19, 23].

Prior work in content-based phone scam detection has mainly utilized machine learning algorithms and Natural Language Processing (NLP) to identify scam content [17, 23], demonstrating that effective fraud detection can be achieved through client-side content analysis. Furthering this research direction and with the emergence of LLM, Zhang et al. [18, 19] developed an approach for phone scam detection that begins by converting live voice streams into text transcripts. These transcripts are then analyzed using LLMs to identify suspicious patterns or fraudulent conversations. When potential scams are detected, the system generates alert messages to notify users in real time. While this approach represents a significant advancement in detecting fraudulent calls, it relies heavily on specific trigger words or phrases rather than a deep understanding of conversational context, making it prone to keyword bias [18]. Such limitation may erode user trust and fairness in real-world deployments of phone scam detection systems. Therefore, while we follow the lead in utilizing LLMs to identify scam phone calls, our system incorporates user evaluations, leveraging real-world feedback to help identify and correct biased or inaccurate detections, thereby improving overall accuracy and adaptability.

3 Scam Detection Design Overview

Our system prioritizes active engagement with end-users in the development of mobile safety tools. It was designed to be customizable, empowering users to decide whether their phone calls should be transcribed and analyzed in real-time. When enabled, the system transcribes calls on the fly and uses LLM models to assess the likelihood of a scam. The models' decision-making process is participatory: users are notified when the model determines that a call is likely a scam and can choose whether to agree with the assessment and decide whether to continue the call or end it. This approach ensures that users remain in control of their communications while benefiting from real-time protection.

- **User-centric design to raise awareness.** End-users are particularly vulnerable to attackers, especially through social engineering techniques that are often sophisticated and difficult to recognize in real-time. To address this, our system was designed not only to detect scam calls but also to actively raise user awareness and support informed decision-making. By delivering real-time notifications whenever a call is flagged as a potential scam, the system helps users understand emerging patterns and tactics that attackers may use to manipulate or deceive them. Over time, this approach strengthens users' familiarity with evolving scam techniques and reinforces their ability to independently identify and respond to potential threats.

- **Transparent and participatory design to protect users.** Anti-Scam was designed with the end-user at its center, ensuring that the entire process is transparent and participatory: no data is scanned, analyzed, or collected without the user's permission, giving users full autonomy over their privacy and communications. Users have the option to opt in or out of enabling our system to scan their phone calls for scam detection. By default, the system operates locally on the user's device, ensuring that personal data never leaves the device unless the user consents. When the system detects a call with a high likelihood of being a scam, it notifies the user and presents its assessment. Users can then choose to agree or disagree with the detection model's evaluation. Importantly, even when a scam is detected, the system does not terminate the call automatically. Instead, it empowers users to make the final decision about whether to continue or end the call. In this way, we have ensured that users benefit from protective measures while maintaining full control over their communications and privacy.
- **Users evaluations for human-in-the-loop mechanism.** As we mentioned earlier, most existing phone scam detection algorithms function as closed systems, offering limited or no opportunity for end-users to evaluate the system's performance. Thus, we designed our system with a built-in, openly accessible model that allow user evaluations of its scam detection accuracy. After each flagged call, users are prompted to confirm whether the model's assessment was correct. With the user's permission, these evaluations are securely collected and integrated into a continuous learning loop that helps refine and improve our model over time. This human-in-the-loop approach not only enhances detection accuracy by incorporating real-world feedback but also fosters transparency and trust, as users see their input directly contributing to the system's development.

4 Technical Implementation

The core implementation of our system required a programming language that supported three key capabilities: real-time audio processing, operating system integration, and Large Language Model (LLM) interoperability, as illustrated in 1. Python was selected as the main programming language for its rich ecosystem of system control libraries and rapid prototyping capabilities.

4.1 Phone Scam Detection System Workflow

A Tkinter-based GUI provided our anti-Scam system with a configurable front-end, allowing users to define detection notification preferences. We leveraged n8n as the workflow orchestrator, connecting various modules to automate and streamline the detection and response process. The Deepgram SDK was employed for real-time speech-to-text transcription, converting live audio streams into analyzable text with low latency. This transcribed text was routed through an AI analysis module that utilized large language models (LLMs) to detect scam-related patterns, phrases, and conversational anomalies. When suspicious content was identified, the system used PyAutoGUI to simulate system-level interactions (i.e., displaying warnings) and Plyer to send immediate cross-platform

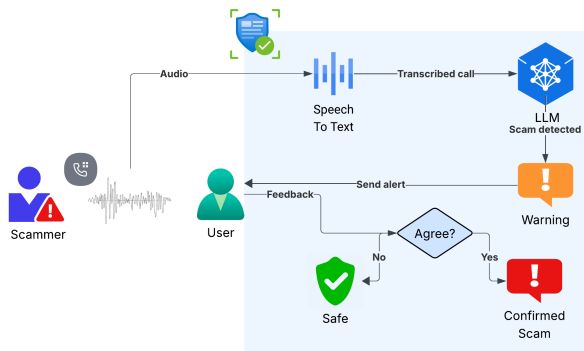


Figure 1: Anti-Scam System Workflow.

notifications to the user, prompting them for feedback on whether they agreed with the LLM-based scam detection decision. We enabled local LLM execution through Ollama, integrated with n8n for workflow automation. The AI agent served as the central link between the LLM and the system’s control tools, incorporating the Model Context Protocol (MCP)—a standard for how applications provide context to LLMs.

4.2 Data Donation Process

For real-time scam detection, we developed a robust system by integrating a local SQLite database node with the AI agent. After the LLM analyzes a phone call and provides a decision, users are prompted to provide feedback on the system’s classification. Their responses, whether they agree or disagree with the detection, are recorded and stored in the local database, accompanied by contextual metadata such as the transcribed call content, timestamp, classification status (e.g., true positive, false positive), and the LLM decision.

For users who consent to contribute their data for future model refinement, we securely synchronize their local databases with a centralized remote server hosted on Amazon Web Services (AWS). This synchronization process begins by packaging the collected data into JSON files, which are then uploaded to Amazon S3 buckets. Once a file arrived in the S3 buckets, an AWS Lambda function is triggered automatically to process and parse the incoming data. The Lambda function extracts relevant information and saves it into an Amazon RDS database. This data is then retrieved and used to fine-tune the models, improving their accuracy based on users’ actual evaluations and labeling of their call conversations.

5 Future Work and Conclusion

Currently, our solution integrates with two major communication platforms—Discord and Zoom—which represent distinct API architectures and voice communication paradigms. This dual integration showcases the system’s adaptability and modularity across both peer-to-peer and client-server models. Thus, we aim to extend the system to support traditional phone call environments, including mobile and VoIP systems, where real-time scam detection is especially critical. This will involve deeper integration with telephony

APIs and OS-level audio routing to ensure seamless transcription and analysis during live calls. Additionally, we have not yet conducted a formal user study, yet we plan to do so to assess the system’s accuracy, usability, and impact on scam prevention in real-world scenarios.

Unlike most AI-based phone scam detection tools that are built privately within corporations, our system offers a transparent and open approach that encourages public involvement. By allowing users to share their feedback on the system’s decisions, we foster a collaborative, community-informed process to enhance the scam detection accuracy. These user contributions and real-world assessments play a crucial role in ensuring the system not only utilizes state-of-the-art LLM models but also provides reliable and user-trusted defense against the growing scam tactics.

References

- [1] Yudhijit Bhattacharjee. 2023. NYTimes: Who’s making all those scam calls? <https://www.nytimes.com/2021/01/27/magazine/scam-call-centers.html>
- [2] Federal Communications Commission. 2023. Caller ID Authentication (STIR/SHAKEN). <https://www.fcc.gov/call-authentication>. Accessed: 2025-05-06.
- [3] Competition and Consumer Protection Commission. 2024. Consumer Protection. <https://www.cccpc.ie/consumers>. Accessed: 2025-05-06.
- [4] Australian Competition and Consumer Commission. 2024. Scamwatch. <https://www.scamwatch.gov.au>. Accessed: 2025-05-06.
- [5] Moya Crockett. 2024. The Secret Health Hell of being scammed: “I felt as though my mind was disintegrating”. <https://www.theguardian.com/society/2024/oct/23/secret-health-hell-being-scammed-felt-mind-disintegrating>
- [6] Federal Bureau of Investigation, Internet Crime Complaint Center (IC3). 2025. *Internet Crime Report 2024*. Technical Report. Federal Bureau of Investigation. https://www.ic3.gov/AnnualReport/Reports/2024_IC3Report.pdf Accessed: 2025-05-06.
- [7] European Union Agency for Cybersecurity (ENISA). 2024. European Cyber Security Month (ECSM). <https://cybersecuritymonth.eu>. Accessed: 2025-05-06.
- [8] Qingchuan Huang, Zhiyuan Zhang, Yanchi Liu, Xun Zhou, Xifeng Yan, Jiawei Han, Thomas W. Yan, Brian LaMaccchia, and Jaideep Srivastava. 2018. Augmenting Telephone Spam Blacklists by Mining Large CDR Datasets. In *Proceedings of the 2018 ACM Asia Conference on Computer and Communications Security (ASIACCS)*. ACM, 711–723. doi:10.1145/3196494.3196553
- [9] Yuxuan Huang, Yan Wang, and Rahul Chatterjee. 2023. Understanding and Detecting Telephone Scams through Voice Phishing Audios. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI ’23)*. Association for Computing Machinery, 1–15. doi:10.1145/3700838.3703672
- [10] Apple Inc. 2024. Detect and block spam phone calls. <https://support.apple.com/en-us/HT207099>. Accessed: 2025-05-06.
- [11] Ibrahim Tariq Javed, Khalifa Toumi, Fares Alharbi, Tiziana Margaria, and Noel Crespi. 2021. Detecting Nuisance Calls over Internet Telephony Using Caller Reputation. *Electronics* 10, 3 (March 2021), 353. doi:10.3390/electronics10030353
- [12] Rasha Kassem. 2024. How fraud impacts individuals’ wellbeing—academic insights and gaps. *Journal of Financial Crime* 31, 5 (2024), 1261–1268.
- [13] Huichen Li, Xiaojun Xu, Chang Liu, Teng Ren, Kun Wu, Xuezhi Cao, Weinan Zhang, Yong Yu, and Dawn Song. 2018. A Machine Learning Approach to Prevent Malicious Calls Over Telephony Networks. In *2018 IEEE Symposium on Security and Privacy (SP)*. 53–69. doi:10.1109/SP.2018.00034
- [14] Google LLC. 2024. Get alerts about suspected scam calls. <https://support.google.com/phoneapp/answer/15654065?hl=en>. Accessed: 2025-05-06.
- [15] Abdul Jabbar Ma’shum and Suharjito. 2020. Fraud Detection Call Detail Record Using Machine Learning in Telecommunications Company. *Advances in Science, Technology and Engineering Systems Journal (ASTESJ)* 5, 4 (2020), 63–69. doi:10.25046/aj050409
- [16] Sharbani Pandit, Mustaque Ahamad, Roberto Perdisci, and Payas Gupta. 2020. Towards Measuring the Effectiveness of Telephony Blacklists. In *Proceedings of the 16th International Conference on Emerging Networking Experiments and Technologies (CoNEXT)*. ACM, 581–594. doi:10.1145/3386367.3431302
- [17] Avula Poojitha, P. Satish Kumar, and Mannuru Malleswari. 2024. Artificial Intelligence Based Fake or Fraud Phone Calls Detection. *International Journal of Novel Research and Development* 9, 7 (July 2024), 319–325. <https://www.ijnrd.org/papers/IJNRD2407033.pdf> Accessed: 2025-05-23.
- [18] Zitong Shen, Kangzhong Wang, Youqian Zhang, Grace Ngai, and Eugene Y. Fu. 2024. Combating Phone Scams with LLM-based Detection: Where Do We Stand? [arXiv:arXiv:2409.11643](https://arxiv.org/abs/2409.11643) [cs.CR]
- [19] Zitong Shen, Sineng Yan, Youqian Zhang, Xiapu Luo, Grace Ngai, and Eugene Yujun Fu. 2025. “It Warned Me Just at the Right Moment”: Exploring LLM-based

- Real-time Detection of Phone Scams. arXiv:arXiv:2502.03964 [cs.HC]
- [20] Daniele Ucci, Roberto Perdisci, Jaewoo Lee, and Mustaque Ahamad. 2020. Towards a Practical Differentially Private Collaborative Phone Blacklisting System. In *Proceedings of the 36th Annual Computer Security Applications Conference (ACSAC '20)*. ACM, 100–115. doi:10.1145/3427228.3427239
- [21] Jian Xing, Miao Yu, Shupeng Wang, Yaru Zhang, and Yu Ding. 2020. Automated Fraudulent Phone Call Recognition through Deep Learning. *Wireless Communications and Mobile Computing* 2020 (2020), 1–9. doi:10.1155/2020/8853468
- [22] Seppo Yrjölä, Marja Matinmikko-Blue, and Petri Ahokangas. 2024. *The Evolution of Mobile Communications*. Springer International Publishing, Cham, 13–43. doi:10.1007/978-3-031-33191-6_2
- [23] Qianqian Zhao, Kai Chen, Tongxin Li, Yi Yang, and XiaoFeng Wang. 2018. Detecting Telecommunication Fraud by Understanding the Contents of a Call. *Cybersecurity* 1, 1 (2018), 8. doi:10.1186/s42400-018-0008-5